HRIDAM ROSHAN PAUL

Prospective Graduate Student

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RESEARCH EXPERIENCES

Experimental and Numerical Investigation of Flexural Energy Absorption in Octopus Suction-Cup-Inspired FDM-Printed Sandwich Structures **(Apr 2025 - Oct 2026)**

- Designed the CAD geometry of the sandwich cores, inspired by octopus sucker morphology, for FDM 3D printing
- Assisted in fabricating specimens by FDM printing and in testing.
- Conducted finite element simulations in Ansys of the sandwich structures and the unit cells.
- Reviewed the literature on bio-inspired and 3D-printed sandwich structures to help identify the research gap, wrote the introduction, contributed to the original manuscript draft, and assisted in addressing reviewer comments.
- Conceptualized and designed the graphical abstract and manuscript visualizations

Synergistic Effects of TiO_2/SiC Hybrid Ceramic Fillers on Jute/Glass Fiber Epoxy Composites **(Oct 2025 - Aug 2026)**

- Identified the research gap on hybrid ceramic filler effects in jute/glass fiber epoxy composites, and designed the study methodology and experimental plan
- Fabricated hybrid jute/glass fiber epoxy laminates with five TiO_2/SiC filler combinations of increasing loading
- Conducted tensile-flexural (UTM), Shore D hardness, density and water absorption tests, and analyzed the data and plotted results in Origin software
- Characterized the microstructure and chemistry by SEM, EDS and FTIR, and correlated the results with mechanical and moisture behavior

Experimental Investigation of Graphite/PLA Composites: Fused Filament Fabrication and FGAM Performance Characterization (Undergraduate Thesis) **(Jul 2026 - Present)**

- Reviewed the literature and defined the research problem: finding the graphite loading that maximizes PLA's thermal stability without compromising filament quality and printability
- Planning to produce PLA/graphite composite filaments at varying wt% and 3D print specimens with varied printing parameters, to study their effect on thermal and mechanical properties
- Planning to fabricate FGAM specimens (uniform, step-gradient and sandwich architectures) and evaluate their thermal and mechanical behavior.

Additive Manufacturing and Sandwich Composite Applications: Book Chapters **(Dec 2025 - June 2026)**

- Co-authored two chapters for *Sandwich Structured Composites: Materials, Design and Applications* (Elsevier), currently under review: "Additive Manufacturing and 3D Printing of Sandwich Structures" (Ch. 11) and "Sandwich Composites in Automotive, Rail, Aerospace, UAV and Marine Applications" (Ch. 29)
- Wrote the sections on design of AM-enabled sandwich cores (11.4), challenges and limitations (11.8) and marine and offshore applications (29.6); managed figure permissions and formatted both chapters.

PROJECT EXPERIENCES

Compact Motorized Multi-Axis Stirring System (*Undergrad Project*) **[See details]** (Dec 2025 - Mar 2026)

- Designed and fabricated an automated stirring prototype integrating CAD modeling, PLA 3D printing, and Arduino Uno based motion control.
- Developed a three-axis mechanism for rotational mixing, vertical positioning, and angular alignment using DC gear motor and servo actuators.
- Implemented manual and automated control modes for sequential multi-container mixing using embedded electronics.

PUBLICATIONS

- S Baidya, S Ahmed, T Ahmed, **HR Paul**, MA Hasib, AM Billah, AK Debnath, "Experimental and Numerical Investigation of Flexural Energy Absorption in Octopus Suction-Cup-Inspired FDM-Printed Sandwich Structures." (Published – Elsevier, ***Materials Today Communications)*** **[Link]**
- **HR Paul**, MJ Rahman, SB Aesop, SMH Aronno, MA Hasib, AM Billah, AK Debnath, "Synergistic Effects of TiO_2/SiC Hybrid Ceramic Fillers on the Mechanical, Morphological and Moisture Resistance Behavior of Jute/Glass Fiber Epoxy Composites." (*Manuscript in Preparation*) **[Manuscript]**
- MA Chowdhury, R Ahammad, **HR Paul**, S Roy, NUH Durjoy, AD Sarker, MM Rahman, "Additive Manufacturing and 3D Printing of Sandwich Structures," Chapter 11 in ***Sandwich Structured Composites: Materials, Design and Applications***, Elsevier. (*Under Review; expected March 2027*). **[Manuscript]**

- MA Chowdhury, R Ahammad, **HR Paul**, S Roy, NUH Durjoy, AD Sarker, MM Rahman, "Sandwich Composites in Automotive, Rail, Aerospace, Unmanned Air Vehicle, and Marine Applications," Chapter 29 in ***Sandwich Structured Composites: Materials, Design and Applications***, Elsevier. (*Under Review; expected March 2027*). **[Manuscript]**

ACADEMIC CREDENTIALS

Bachelor of Science in Mechanical Engineering **Jan 2023 – May 2027 (Expected)**
Khulna University of Engineering & Technology (KUET), Bangladesh

Cumulative GPA: **3.23**/4.00 (up to 6th semester)

THESIS (ongoing)

Experimental Investigation of Graphite/PLA Composites: Fused Filament Fabrication and FGAM Performance Characterization.
Supervisor: Prof. Dr. Md. Shariful Islam, Department of Mechanical Engineering, KUET

- Investigating the graphite loading that gives PLA the best thermal stability, along with filament quality, printability and FGAM performance, in FFF-printed specimens

HONORS AND AWARDS

- **2nd Runner Up, National CAD Contest – Ignition 2026**, organized by the Dept. of Mechanical Engineering, KUET (May 2026)
- **1st Runner Up, National CAD Contest – TECHNOMIZE 1.0**, organized by IEM Robotics & CAD Club, KUET (Nov 2024)
- **1st Runner Up, National CAD Contest – Calibration 1.0**, organized by the Dept. of Mechatronics Engineering, KUET (Feb 2024)
- **Technical Scholarship recipient**, Khulna University of Engineering & Technology (KUET): awarded for the 1st-year and 2nd-year academic sessions.

SKILL SUMMARY

- **CAD, Design & Simulation:** Ansys (static structural, fluent), SolidWorks, AutoCAD, Abaqus.
- **Materials Testing & Characterization:** Universal Testing Machine (Tensile/Flexural), Shore D Hardness, SEM, EDS, EDX, FTIR, Water Absorption
- **Fabrication:** FDM/FFF 3D Printing, Filament Extrusion, Hand Lay-up.
- **Data Analysis & Modeling:** Origin Pro.
- **Programming Language:** Python, C.
- **Documentation & Digital Tools:** Microsoft Office, Google Workspace, Canva, WordPress, Adobe Premiere Pro, Technical Writing.

CERTIFICATIONS

- Certified SOLIDWORKS Professional (CSWP), Dassault Systèmes, 2024 **[Credential]**
- Certified SOLIDWORKS Associate – Mechanical Design (CSWA), Dassault Systèmes, 2024 **[Credential]**
- Technical Drawing & CAD Competition, CADers, 2023 **[Credential]**
- Professional Content Writing, BrightSkills, 2022 **[Credential]**

TEACHING AND MENTORING EXPERIENCE

CADers, KUET [Official page] **Dec 2024 – Mar 2026**
Instructor & Mentor (Computer aided design workshops)

- Delivered offline AutoCAD and SolidWorks workshops for undergraduate engineering students, with 40–50 participants per session and classes of nearly three hours.
- Prepared lecture content, presentations, and classwork and homework in advance, and demonstrated the software live while teaching.
- Mentored students preparing for national-level CAD contests and international certification exams.

10 Minute School **(The largest online education platform in Bangladesh) [*Website*]** **Aug 2023 – Jul 2025**
Junior Teacher - Doubt Solver (Contract)

- Resolved students' academic doubts in physics, chemistry, biology, and mathematics during and after live classes in 10 Minute School's academic programs, providing online guidance to students in grades 6–12, including O Level and A Level candidates.

LEADERSHIP & TECHNICAL EXPERIENCE

IMechE KUET Student Chapter [Official page] **Aug 2026 – Present**
Policy Analyst

- Plan chapter events, prepare formal documentation and manage abstract submissions and participant registration for the SOfE (Speak Out for Engineers) heat.

CADers (Computer Aided Designers Club) [Official page] **Mar 2024 – Present**

Quality Audit Officer (Aug 2026 - Present)

- Oversee quality, compliance and operational standards of club activities as part of the independent Audit Board, conducting periodic audits under the club constitution and reporting to the Faculty Advisor.

Chief Executive Officer (Jan 2025 – Aug 2026)

- Led a working committee of 40 members across 6 sub-teams, and collaborated with clubs and student organizations across Bangladesh on technical and community initiatives.
- Organized SolidWorks and AutoCAD workshops for nearly 200 participants, supported by continuous evaluation and intra-club contests, along with CAD contests, technical events and guideline sessions for undergraduates.

Assistant IT & Resource Officer (Mar 2024 – Jan 2025)

- Assisted instructors in workshops by supporting participants one-on-one and giving feedback on submitted homework.
- Produced YouTube tutorials with screen recordings and voice-over in Adobe Premiere Pro; prepared sponsorship proposals, approached local tech companies and designed graphics for the club.

Team Kilo Flight (FSAE team) [Official page] **Mar 2024 – Aug 2026**

Senior Executive – Body & Aerodynamics sub-team (Dec 2024-Aug 2026)

- Designed the hub, ran structural simulation to visualize stress and strain, and optimized the design.
- Assembled CAD models of the steering and braking systems; interviewed new trainee members for the sub-team.

Trainee member – Body & Aerodynamics sub-team (Mar 2024-Dec 2024)

- Learned and practiced airfoil and static structural simulation in Ansys Workbench; studied topics for FSAE competition quizzes.

INTERNSHIP EXPERIENCE

Youth School for Social Entrepreneurs (YSSE**: youth-led non-governmental organization) [LinkedIn]** **Oct 2022 – Aug 2023**

Content Writing Intern (Oct 2022 – Mar 2023) → Management Trainee (Mar – Jun 2023) → Junior Associate (Jun – Aug 2023)

- Wrote blogs for youth readers that were published on the YSSE blog site, and proofread colleagues' drafts before final approval.
- Managed a group of content-writing interns: approved blog captions, assigned proofreading tasks, tracked deadlines and hosted online meetings.
- Collected updates from Management Trainees and reported to the department head; coordinated with the YSSE President to request approval for sessions and campaigns; interviewed intern applicants, published selected blogs on the WordPress site and kept in contact with other departments.

VOLUNTEER AND EXTRACURRICULAR ACTIVITIES

- **Volunteer, Hult Prize at KUET 2024** (Jan – Feb 2024): assisted in event coordination and operations during the competition.
- **Executive Committee Member, Josephite Math Club, Dhaka** (Aug 2019 – Dec 2021): assisted in conducting Math Olympiads and other competition segments for students up to higher secondary level at the Josephite Math Mania 2019.

REFERENCES

Dr. Mohammad Ilias Inam
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